Degradation-Aware Dispatch Optimization for Residential Vehicle-to-Building (V2B) and PV Systems

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***Abstract*—Vehicle-to-Building (V2B) charging integration presents a promising pathway to reduce residential electricity costs and grid strain while maximizing rooftop solar photovoltaic (PV) consumption. However, prior V2B models typically neglect detailed battery degradation modeling which presents a critical gap given that unregulated bidirectional charging can accelerate electric vehicle (EV) battery degradation from increased cycling, potentially eliminating the economic gains of V2B. This study evaluates how incorporating a battery degradation penalty into a V2B optimization model impacts dispatch strategies, economic outcomes, and battery wear in residential V2B systems.**

To achieve this, a mixed-integer linear programming (MILP) model was developed to optimize the hourly summer and winter dispatch of a California residential home with a 6 kW PV array and a 60 kWh EV battery under a Time-of-Use pricing structure. We compare degradation-naive V2B and degradation-aware V2B strategies. To accurately measure battery wear, the state-of-charge (SoC) profiles are analyzed afterwards using rainflow cycle counting to quantify battery degradation.

By varying the degradation penalty weight, a Pareto tradeoff curve is generated to quantify the tradeoff between short-term utility bill savings and long-term battery degradation cost. Ultimately, this research provides quantitative insight into the economic tradeoffs of residential V2B implementation, balancing energy cost savings with battery health instead of ignoring the losses incurred by degradation.

Keywords—battery, photovoltaic, vehicle-to-building, EV

I. Introduction

The transition to renewable energy has positioned the modern electric vehicle (EV) as more than just a mode of transportation; it is increasingly viewed as a mobile, high-capacity energy storage system. Vehicle-to-Building (V2B) technology enables an electric vehicle to act as a battery, charging from the grid and discharging stored energy to a home in times of peak grid load. When paired with residential renewable energy sources, V2B charging presents a promising pathway to reduce residential electricity costs and grid strain while maximizing rooftop solar photovoltaic (PV) consumption [3].

The primary financial driver for adopting residential V2B systems is the manipulation of Time-of-Use (TOU) electricity pricing. Under TOU structures, utility companies charge higher rates during periods of high demand (typically late afternoon and evening) and lower rates during off-peak hours or when solar generation is abundant. By combining V2B with rooftop solar (PV) and TOU pricing, homeowners can store cheap solar or off-peak grid energy and discharge it during expensive peak hours to lower electricity bills and relieve grid pressure. In essence, the homeowner uses the EV battery to "buy low" and "consume high," insulating themselves from the most expensive grid rates.

Despite the apparent financial benefits, a significant blind spot in current V2B research is the lack of attention given to battery wear in previous models [2, 4]. This neglect of detailed battery degradation modeling presents a critical gap given that unregulated bidirectional charging can accelerate electric vehicle (EV) battery degradation from increased cycling [1]. Similar to the gradual loss in capacity of smartphone batteries after years of daily charging and discharging, an EV battery experiences microscopic physical and chemical wear with every cycle. Forcing an EV to power a home every single day exacerbates this stress and if not accounted for, could lead to unexpected battery replacement costs, potentially negating the economic gains of V2B.

II. Overview

To address the oversight of EV battery degradation, this project aims to find the tradeoff between maximizing savings and preserving battery health. The research evaluates how incorporating a battery degradation penalty into a V2B optimization model impacts dispatch strategies, economic outcomes, and battery wear in residential V2B systems. Ultimately, this analysis provides quantitative insight into the economic tradeoffs of residential V2B implementation, balancing energy cost savings with battery health instead of ignoring the losses incurred by degradation.

III. Methodology

To understand how battery wear impacts the financial viability of Vehicle-to-Building (V2B) systems, this study developed a computational model to simulate a residential home's energy usage. The goal was to mathematically determine the most cost-effective way to route power between the grid, the solar panels, the home, and the EV battery on an hour-by-hour basis.

The simulation was modeled around a typical residential home located in Pasadena, California. Pasadena was specifically selected because it offers an ideal environment for testing V2B dynamics with its high solar irradiance for

reliable power generation and a moderately aggressive Time-of-Use (TOU) pricing structure. As seen in Fig. 1, the virtual home was equipped with a standard-sized 6 kW rooftop PV array capable of significantly offsetting a home's daily energy needs and a 60 kWh EV battery typical of standard-range modern electric vehicles. In the simulation, the car leaves at 8 AM, returns at 6 PM, and consumes roughly 10 kWh of energy while away. There is a 9.6 kW bidirectional charger and state-of-charge (SoC) bounds for the battery are capped between 20% and 90% to protect the battery from deep discharge damage and high voltage stress.

The simulation mainly relies on a Mixed-Integer Linear Program (MILP), built using the Pyomo [7] modeling language and solved with Gurobi [8]. The simulation is effectively an advanced scheduling algorithm with the goal of optimizing the flow of electricity to minimize cost to the homeowner as represented by the objective function (1). Provided with the hourly TOU electricity rates from Southern California Edison (SCE) [9], the household's power consumption data from NREL OpenEI [10], and the local solar generation data from PVWatts [11], the model determines the optimal hourly "dispatch" (routing) of energy for typical summer and winter days.

$$\min \sum_{t=1}^{24} \Big[\underbrace{\pi_t^{\text{buy}} \cdot P_t^{\text{buy}}}_{\text{grid cost}} - \underbrace{\pi_t^{\text{sell}} \cdot P_t^{\text{sell}}}_{\text{export credit}} + \underbrace{w \cdot (P_t^{\text{ch}} + P_t^{\text{dis}})}_{\text{degradation penalty}} \Big] \tag{1}$$

To isolate the impact of battery degradation, the optimization model was run under three distinct operational modes. In Mode A, the EV functions purely as a consumer, charging when electricity is cheapest or solar is abundant. This mode acts as a baseline because the battery cannot discharge power back into the home. In Mode B, the system is allowed to charge and discharge the EV battery freely with the degradation penalty weight (w) being set to 0. The model's only goal in this mode is to maximize financial savings by exploiting the TOU pricing spread, disregarding any battery wear from aggressive cycling. Lastly, in Mode C, a nonzero linear battery penalty weight (w) is introduced as a constraint in the objective function (1). The optimization model is charged a hypothetical "tax" for every kilowatt-hour of energy going through the battery (both charge and discharge) to estimate the financial cost of EV battery degradation. The model must therefore perform a balancing act, only discharging the battery to power the home when grid prices are high enough to justify the "cost" of battery wear.

TABLE I. SUMMARY OF MODES

Mode	Description	Degradation Penalty (w)
A	EV can only charge from the grid or PV. V2B discharging is disabled.	None (w constraint ignored)
B	EV charges and discharges freely to maximize financial TOU arbitrage.	$w = 0$ (Free to cycle)
C	Optimization factors in battery wear for every kWh charged or discharged.	$w > 0$ (Cycling incurs a cost)

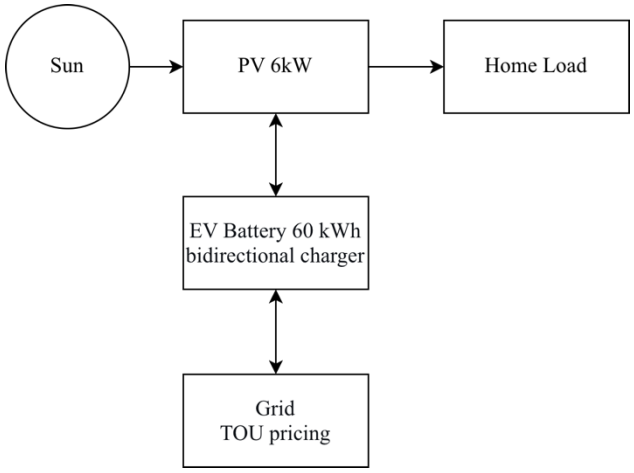


Fig. 1. System architecture detailing the main components of the simulation

True battery degradation is more complex however, and does not follow a linear penalty. For example, depth of discharge plays a large role. Discharging a battery from 30% down to 10% causes significantly more damage than discharging that same 20% from 90% down to 70%. Because the optimization model (Mode C) used a simplified linear penalty to estimate the battery cost, a more accurate method to verify the battery damage actually being done was necessary. To quantify the true stress caused V2B use while factoring in the depth of discharge, the battery's state-of-charge (SoC) profiles were analyzed post-hoc using a technique called rainflow cycle counting.

Originally developed to measure metal fatigue in aerospace and civil engineering, the rainflow counting algorithm enabled more complex analysis of the stress cycles by taking into account increased damage from depth of discharge [5, 6]. This way, the true wear on the EV battery could be measured more accurately under the different V2B strategies.

IV. RESULTS AND FINDINGS

A. Seasonal Economic Viability of V2B

Analysis of the financial outcomes from the V2B simulation clearly indicate that the economic viability of V2B is highly dependent on seasonal TOU spreads. Ultimately, typical TOU pricing structures do not, by default, incentivize year-round V2B usage. As seen in Table II, the system performs best during the summer months, where utilizing V2B yields an 11% cost reduction from the baseline which is roughly $1.53 in savings per day. This success is largely driven by wider price gaps during summer peak hours and abundant solar generation. Conversely, winter performance is drastically lower with winter V2B having only a 3.3% cost reduction amounting to a nearly negligible $0.33 in savings per day. The simulation results indicate that without an additional financial incentive, utilizing V2B to cycle an EV battery daily during the winter provides little benefit to the homeowner.

TABLE II. SUMMARY OF RESULTS

Metric	Summer Mode A (baseline)	Summer Mode C (w = 0.04)	Winter Mode A (baseline)	Winter Mode C (w = 0.04)
Daily Electricity Cost	$12.90	$11.37	$10.01	$9.68
Peak TOU (per kWh)	$0.58	$0.58	$0.51	$0.51
Total Load (kWh)	38.64	38.64	24.42	24.42
PV Output (kWh)	33.46	33.46	20.38	20.38
V2B Energy Discharged (kWh)	0.00	7.58	0.00	3.26
Rainflow Degradation Index	0.027494	0.067399	0.027778	0.050151

B. The "Binary Behavior" Anomaly and Linear Modeling Flaws

A major objective of this study was to identify a "sweet spot" balancing utility savings with battery health. The original expectation was that by varying the degradation penalty weight, a smooth Pareto curve could be generated to quantify the tradeoff between short-term utility bill savings and long-term battery degradation cost. However, as seen in Fig. 5 and Fig. 6, applying a linear throughput penalty results in binary behavior rather than a smooth, gradual tradeoff. Instead of incrementally reducing the battery's discharge depth as the penalty increased, the optimization model acted like an on/off switch. The system cycles aggressively right up until the penalty weight becomes larger than the price difference between peak and off-peak electricity. This is the moment that the "degradation tax" costs more than the TOU arbitrage profits and the model will instantly shut down all V2B activity to save money. This results in a sudden jump from heavy degradation directly to the mode A baseline with no vehicle discharge. This anomaly exposes a critical modeling shortcoming in that a simple linear penalty lacks the nuance to produce partial-dispatch V2B strategies that would be more representative of real-world implementation.

Additionally, when verifying the true stress on the battery using post-hoc cycle counting, the rainflow analysis showed that the linear model severely underestimated the true cost of aggressive V2B cycling. Because the standard linear optimization treated all energy throughput equally, it failed to recognize that the deeper cycles caused exponentially more damage than shallow ones. As a result, the actual battery degradation observed through rainflow analysis can be 2.5 to 3.5 times higher compared to the observations of the optimization model in Mode C, a sizable hidden cost largely invisible to standard linear optimization.

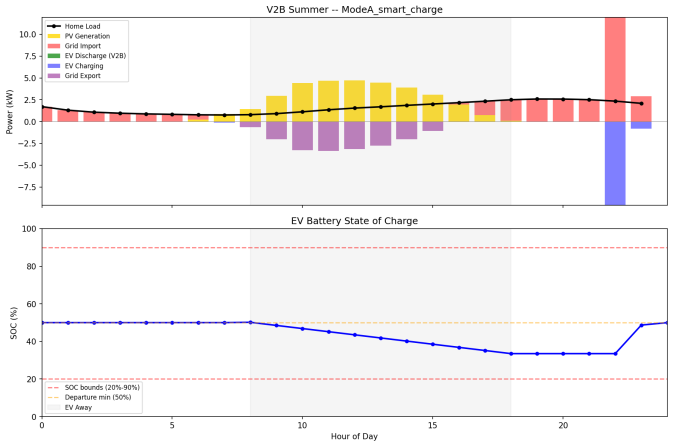


Fig. 2. Summer Mode A dispatch and corresponding battery SoC

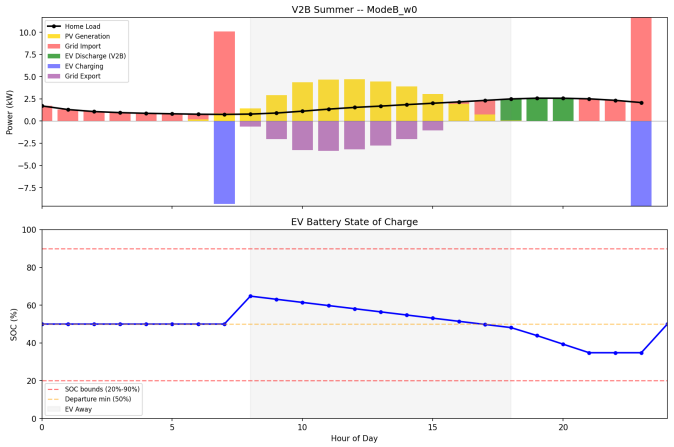


Fig. 3. Summer Mode B dispatch and corresponding battery SoC

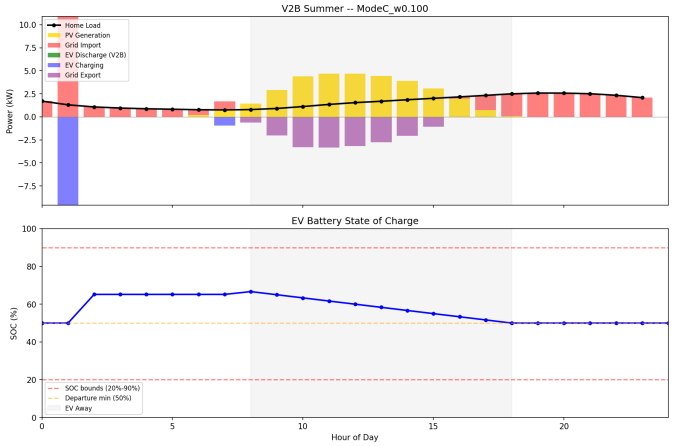


Fig. 4. Summer Mode C dispatch and corresponding battery SoC

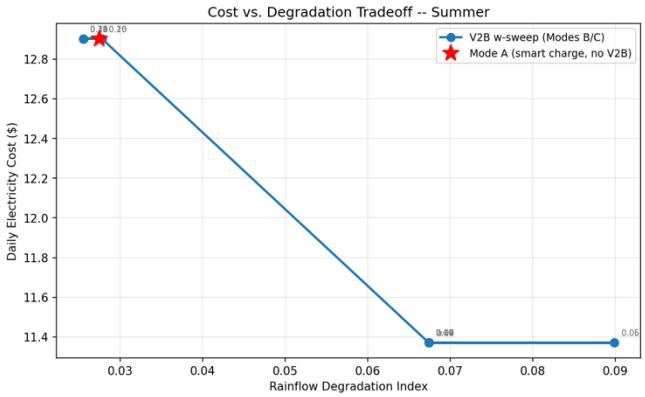


Fig. 5. Summer tradeoff curve showcasing binary behavior. The degradation jumps from heavy degradation to Mode A baseline.

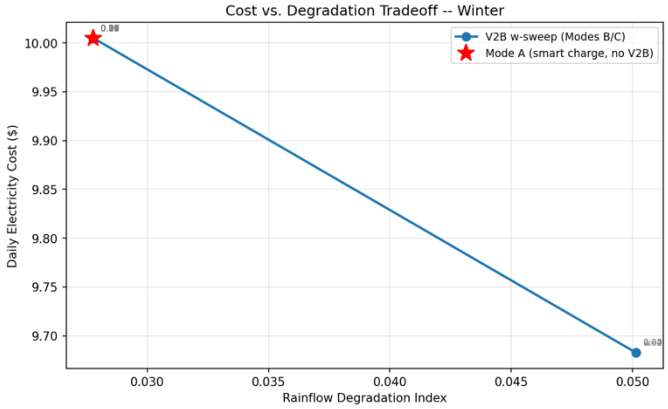


Fig. 6. Winter tradeoff curve showcasing binary behavior.

V. Conclusion and Next Steps

This study evaluated how incorporating a battery degradation penalty into a V2B optimization model impacts dispatch strategies and economic outcomes for residential solar-integrated homes. By utilizing a Mixed-Integer Linear Programming (MILP) model to simulate a California residence, the research compared the performance of degradation-naive and degradation-aware strategies across the winter and summer seasons. The findings indicate that the financial viability of V2B is heavily dependent on seasonal Time-of-Use (TOU) spreads, with significantly lower cost savings in the winter which may not justify the added stress on the EV battery. Crucially, the research identified significant modeling shortcomings regarding linear degradation penalties. Instead of a smooth Pareto tradeoff between savings and battery health, the linear throughput penalty created a "binary behavior" anomaly, where the system acted as an on/off switch for V2B activity. Furthermore, post-hoc rainflow cycle counting revealed that the linear model underestimated true battery degradation because it failed to account for the exponential damage caused by deeper discharge cycles. To address these gaps, future V2B optimization models must move beyond simple linear "taxes" on energy throughput, instead integrating nonlinear, cycle-depth-aware degradation penalties directly into the dispatch logic. Subsequent research should also consider exploring more complex Time-of-Use structures that might better incentivize year-round V2B participation and adding a second home battery for greater energy storage.

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